Federico Cesarini

AI Engineer & Consultant · Solo Founder, SecondBrain

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PROFILE

AI engineer and consultant. I build memory systems for agentic AI and, more broadly, generative AI products that run in production, from the agents people talk to down to the cloud infrastructure underneath. At Tinexta I lead the engineering of the main AI features of Lextel.ai, a platform for legal professionals: deep research, document generation and an agentic legal chat. On the memory side, I built SecondBrain, an AI memory app with paying subscribers on the App Store and Google Play, and, together with Marco Sassarini, CROW, a paper and open-source project that makes knowledge bases maintained by language models cheaper and more accurate.

RESEARCH & OPEN SOURCE

CROW: Classifier-Routed Organization of LLM Wikis

Sep 2026

First author, with Marco Sassarini · paper, open-source implementation and desktop app

Paper doi.org/10.5281/zenodo.23260209 | Code github.com/crow-hq/llm-wiki-v2 | Desktop app github.com/crow-hq/crow-desktop

- In an LLM wiki, a language model files documents into a tree of notes and answers questions from them, but every small decision, such as where a document belongs or which note to read, costs a model call. CROW hands those decisions to a fast, cheap classifier and keeps the language model only for writing.
- On 525 real documents, compared with the same pipeline driven by a language model, CROW added documents at half the cost and in half the time (18 s instead of 39 s each), and found the right note for a question 89% of the time instead of 62%, about twice as fast.
- The cost of each new document stays flat as the wiki grows, and a cost model built on these measurements keeps it at about $0.006 per document up to a million documents.
- The open-source implementation, which I maintain, is a wiki of plain Markdown files that runs on your own computer with any AI provider and cites its sources (200+ stars and 30+ forks on GitHub); Crow, its desktop app for macOS, Windows and Linux, is made with Tauri and React.

CORE COMPETENCIES

AI & Agents	Memory for agents, retrieval-augmented (RAG) systems and agents that call tools and connect to other apps through MCP, along with deploying and fine-tuning both large and small language models, from Gemini and GPT to open-source ones, with frameworks such as the OpenAI Agents SDK, Google ADK and SmolAgents.
Cloud & Data	Most production work runs on Google Cloud, with Kubernetes that scales with the workload, Cloud Run and Cloud Functions, and vLLM to serve models, alongside Elasticsearch clusters for keyword and semantic search; my own products run on Supabase.
Engineering	Python with Django and FastAPI on the backend, TypeScript with React Native and Expo for mobile apps, Celery, RabbitMQ and Redis for background work, Docker for packaging, Prometheus and Grafana for monitoring, and the full release cycle on the App Store and Google Play.
Growth	Making ads and running campaigns on Instagram and Facebook, keeping apps easy to find in the stores, on Google and in AI assistants, and deciding where to spend by comparing what a paying user costs with what they bring in.

FLAGSHIP PRODUCT

SecondBrain · AI memory app for iOS, Android and web

2026 – Present

Solo founder and builder: engineering, product, design and growth

secondbrainmemory.com | MCP github.com/fed3c3sa/secondbrain-shared-memory | App Store apps.apple.com/it/app/secondbrain-note-vocali-ai/id6762130376 | Google Play play.google.com/store/apps/details?id=app.secondbrain.android

- Designed, built and launched as a one-person project, first on the App Store and then on Google Play; it is live on both today, with paying subscribers.
- You talk to it and it writes a titled note, files it in the right folder and turns what you said into calendar events and reminders; voice messages shared from WhatsApp work the same way.
- It also gives Claude, ChatGPT, Gemini and Cursor one shared memory built on your own notes, through an MCP connector that is available as a Claude Code plugin too.
- The backend runs on Supabase, with an AI agent that shows its progress live, switches to another model when one fails so no note is lost, and keeps working offline.
- More than 15 updates so far, up to version 2.8.0, in five languages and with markdown export so users can always take their notes with them; onboarding, free trials and the paywall were reworked around how users actually behaved.

GROWTH & MARKETING

Founder-led growth · SecondBrain

Jun 2026 – Present

All of the marketing, from making the ads to deciding where the budget goes

- More than 150 ads made so far, from creator-style videos and graphics to short vertical videos.
- Paid campaigns on Instagram and Facebook, steered by a self-built dashboard that puts ad spend, subscriptions and payments side by side, to see what each new paying user costs and keep only the ads that pay for themselves.
- Growth without paid ads as well: running the app's Instagram account, making the website easy to find on Google and in AI assistants such as ChatGPT, improving the store listings and taking part in Reddit communities.

PROFESSIONAL EXPERIENCE

AI Engineer & Cloud Architect

Feb 2025 – Present

Tinexta Innovation Hub · Lextel.ai

- Led the development of the main AI features of Lextel.ai: Legal Deep Research, AI document generation and an agentic legal chat.
- Redesigned the document ingestion pipeline on Kubernetes so that it scales with the workload, doubling its throughput, and added Celery to process large volumes of data in parallel.
- Built the retrieval (RAG) pipelines that let the AI answer from legal and patent documents, including the Kubernetes pipeline that splits, embeds and indexes them, monitored with Grafana and Prometheus.
- Ran the Elasticsearch clusters and tuned their queries, deployed machine-learning models inside them with Eland and benchmarked them with ES Rally.
- Mentored junior developers and helped them become productive faster.

Software Engineer & Cloud Architect

Dec 2021 – Feb 2025

Warrant Hub · Bergamo

- Built the Google Cloud infrastructure used to process patent data at scale.
- Trained named-entity recognition models that extract material information from patents with 85% accuracy, and built the pipeline that reads unstructured patent text.
- Designed the Elasticsearch databases that make patents searchable both by keyword and by meaning.
- Built Django backends and Kubernetes pipelines that automate data processing.

Software Developer
Jun 2020 – Oct 2020

GestApp srl · Bergamo

- Wrote C# and VB.NET backend code and business logic for enterprise applications.

Retail Employee
Aug 2019 – Jan 2020

Apple Inc. · customer service and technical support in an Apple Store

OTHER PROJECTS

Elasticsearch Natural Language Query github.com/fed3c3sa/nlq_translator
A service that lets you question Elasticsearch in plain language: GPT writes the query and refines it, using the structure of the index as context.

linkAIin, LinkedIn AI Auto-Poster github.com/fed3c3sa/linkAIin
A Google Cloud Function that researches, writes and publishes LinkedIn posts with GPT, optionally with DALL-E images, and tracks how they perform.

PUBLICATIONS

CROW: Classifier-Routed Organization of LLM Wikis — Sep 2026
with M. Sassarini, Zenodo preprint

NER in the LLM Era: How I Used Giant Models to Train Tiny Ones — Dec 2025

How to Deploy Any Open-Source LLM on Google Cloud Without Owning a Single GPU — Nov 2025

The AI Memory Problem: Why RAG Needs to Evolve Into Something More — Nov 2025

How to Deploy an SLM on Cloud Run and Scale to Zero — Oct 2025

Articles on Medium medium.com/@mail.fede.cesarini

EDUCATION & LANGUAGES

Bachelor of Science in Computer Engineering
Sep 2017 – Aug 2021

Università degli Studi di Bergamo · Bergamo

Languages. Italian (native) · English (C1)

